Heegner-Golden Figure Atlas

Twelve reproducible mathematical figure plates

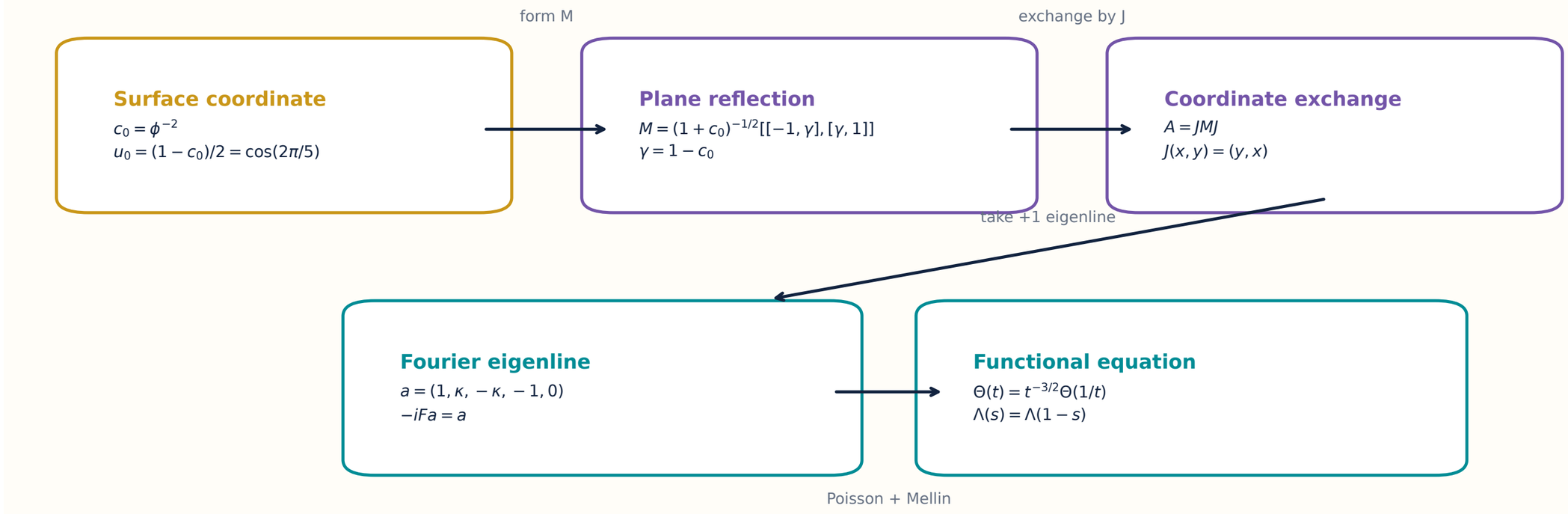
Native vector artwork; numerical arrays and high-precision diagnostics are saved with the packet.

Colour key: gold geometry | teal Fourier/theta | violet modular map | blue CM/Hermite

Ruled-surface geometry, Rogers-Ramanujan inversion, and Fourier self-duality.

Figure 01. Connection map

Figure 01. One scalar coordinate, an explicit involution, and its Fourier consequence



Distinct CM branch: i is fixed by $S : \tau \mapsto -1/\tau$ and selects $\kappa = r(i)$.
$\tau_{163} = (1 + i\sqrt{163})/2$ instead supplies $j(\tau_{163}) = -640320^3$ and a tiny negative nome.

Golden geometry = gold | Fourier / theta = teal | modular map = violet | CM / Hermite = blue

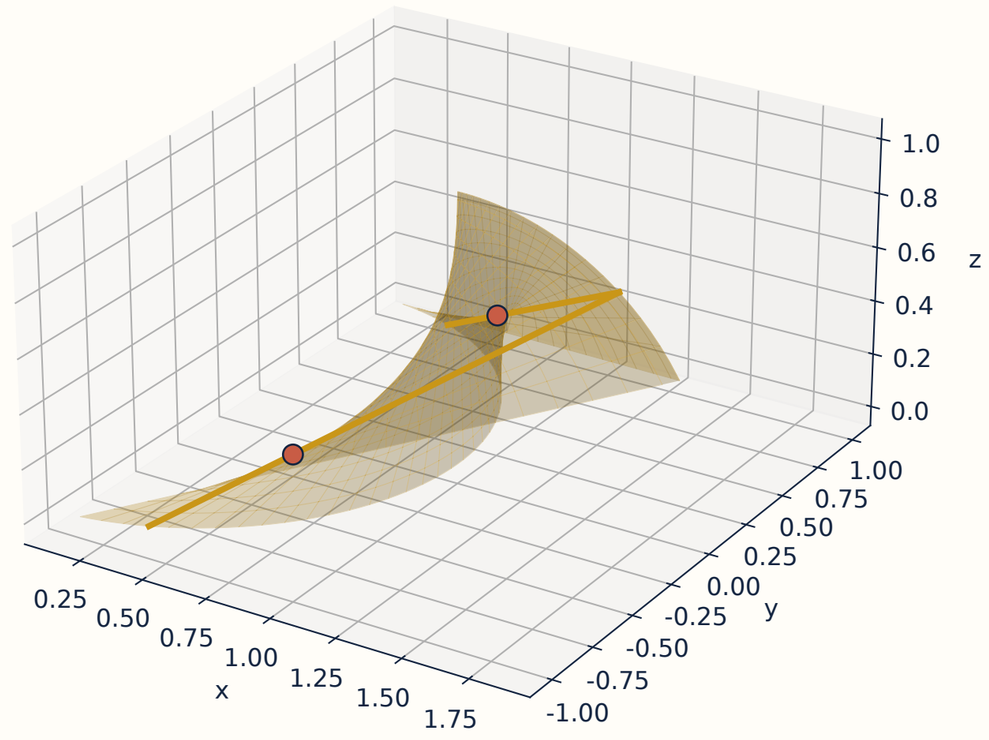
A navigation map: the golden surface coordinate determines an explicit reflection; coordinate exchange leads to the odd Fourier eigenline and its functional equation. The i and tau_163 CM roles remain distinct.

Placement cue: Opening orientation after the scalar definitions.

Figure 02. The orthogonal-circle ruled surface and its two lateral rank-loss points

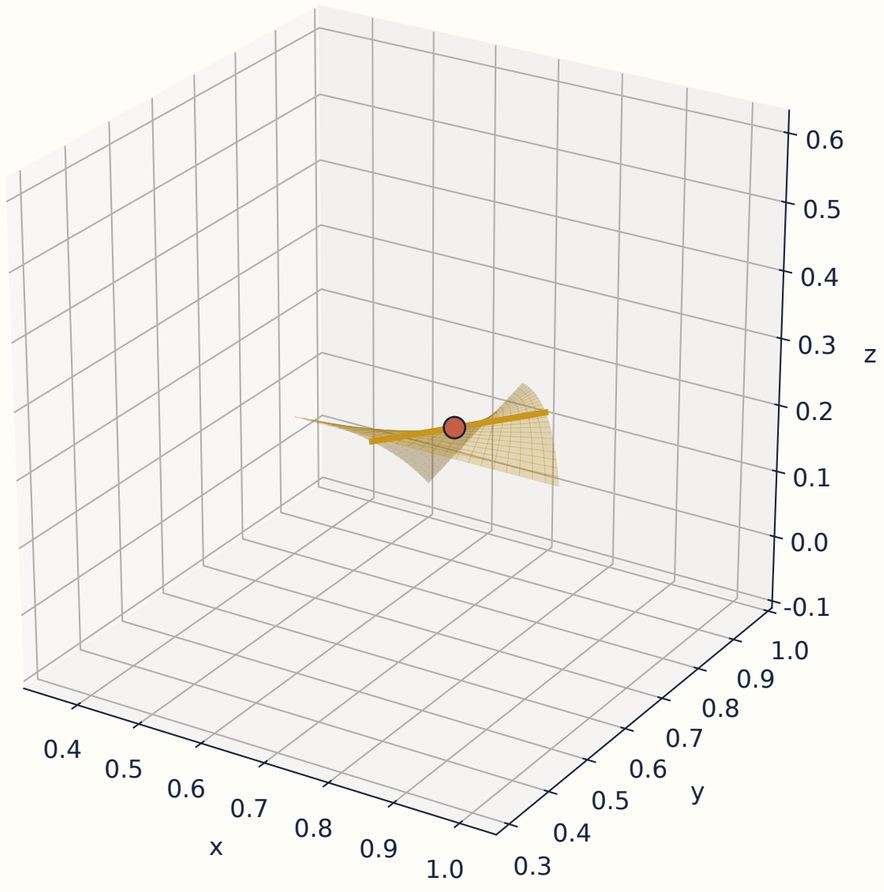
Actual ruled patch $S(t, u)$

Coral markers: the lateral pair



Local patch at the positive-t lateral point

$p_+ = S(\arccos c_0, u_0)$

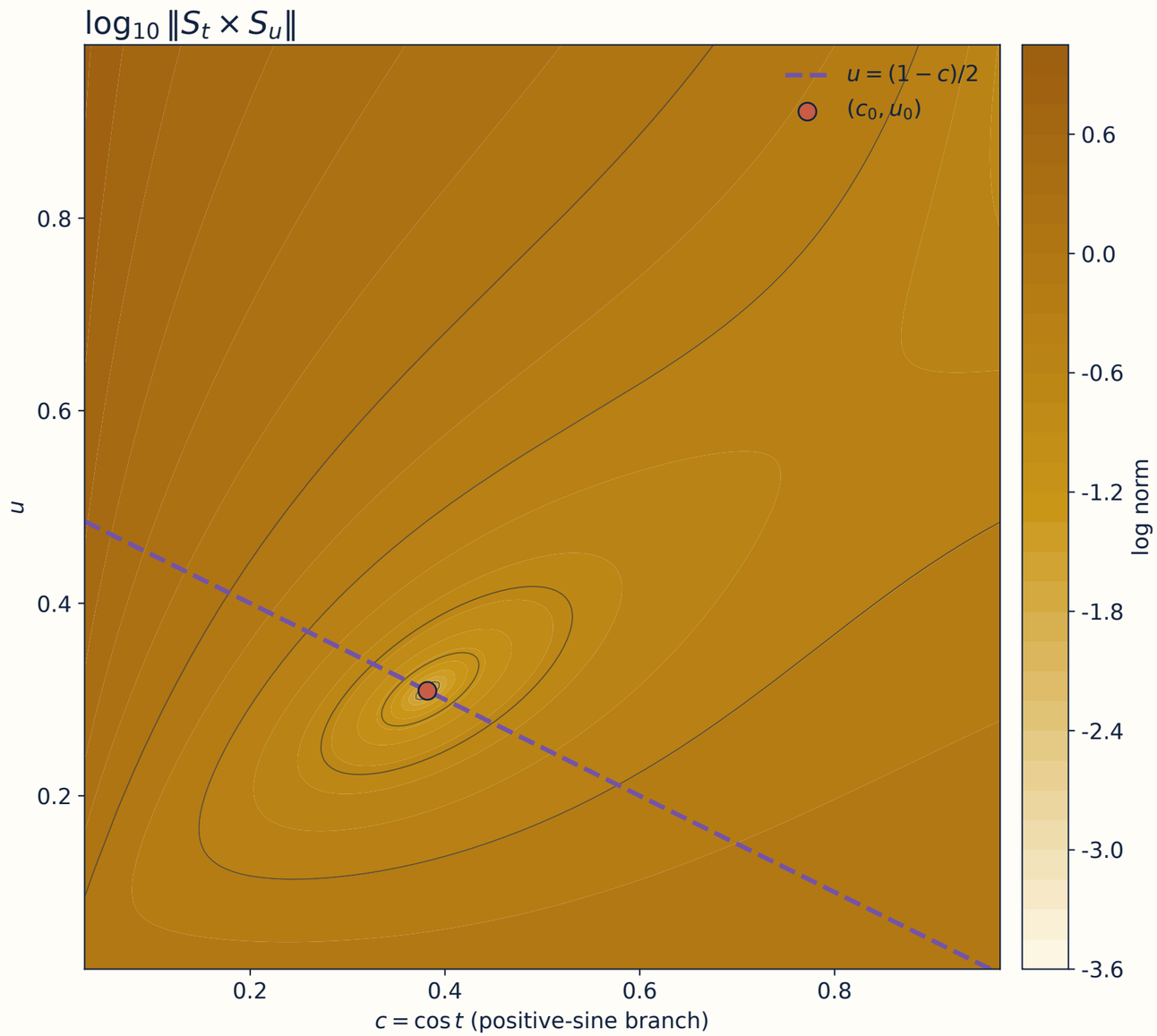


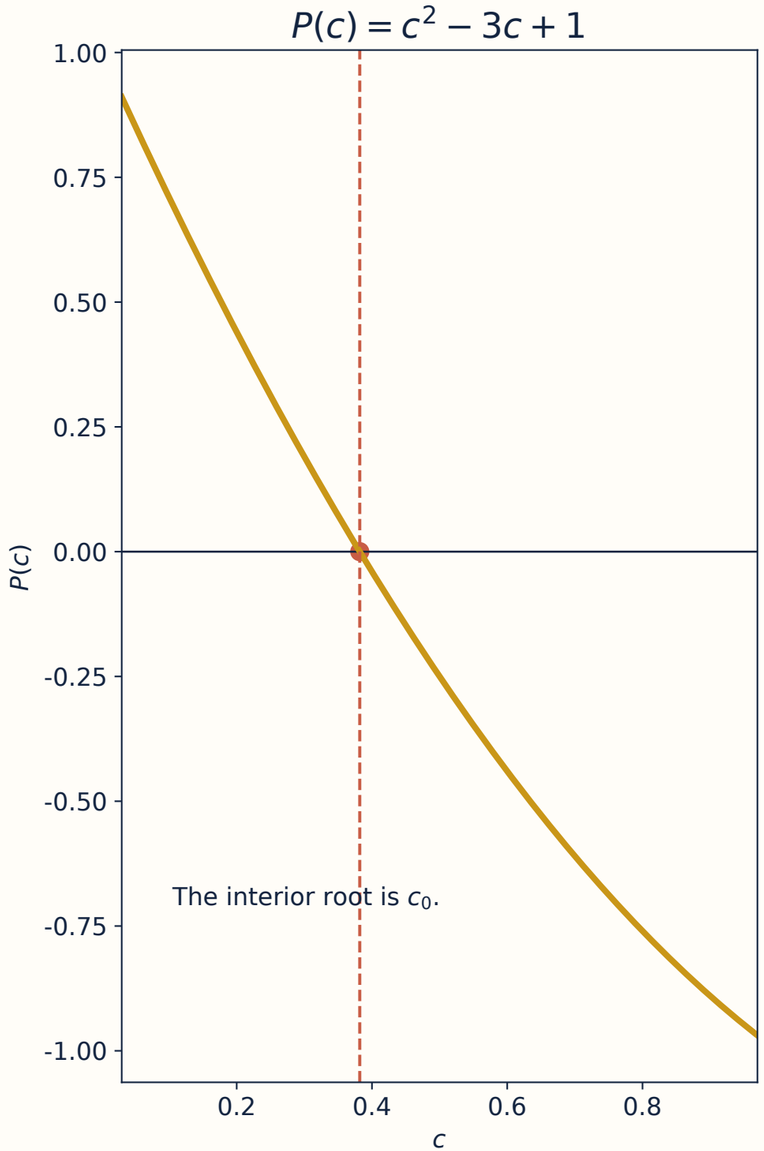
Golden geometry = gold | Fourier / theta = teal | modular map = violet | CM / Hermite = blue

The actual ruled patch S(t,u), with its reflected lateral rank-loss pair. The right panel is a local view of the positive-t point, using equal coordinate spans.

Placement cue: Section 1, after the ruled-surface parametrization.

Figure 03. Rank loss on the positive-sine branch: the lateral condition and its root





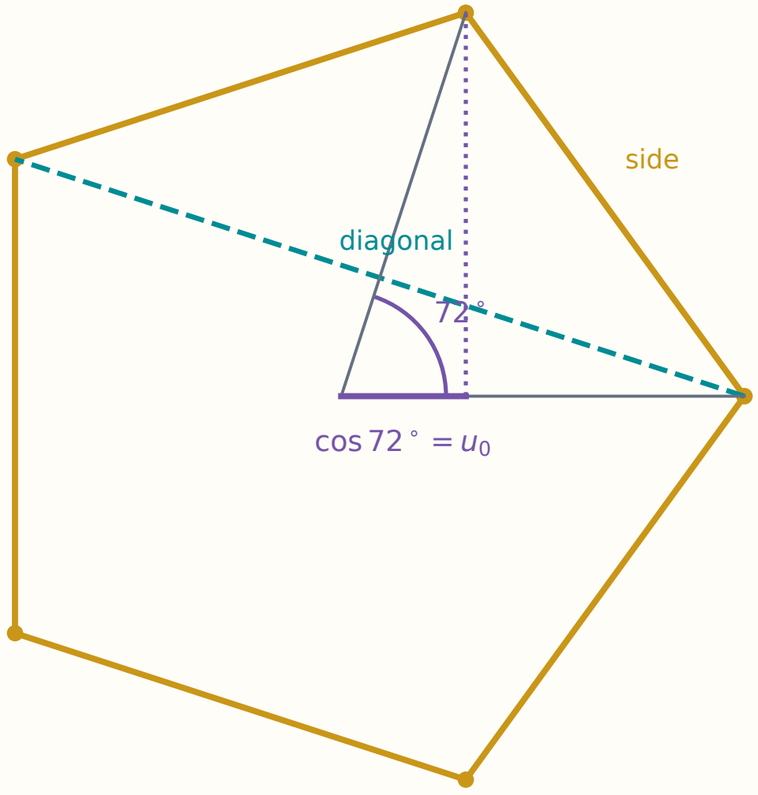
Golden geometry = gold | Fourier / theta = teal | modular map = violet | CM / Hermite = blue

On the positive-sine branch, the norm of the actual cross product vanishes at the lateral condition; the adjacent polynomial isolates its interior root c0.

Placement cue: Section 1, beside the rank-loss elimination.

Figure 04. The pentagonal coordinate: angle, projection, and the golden squared ratio

Unit pentagon: vertices 0, 72, 144, ... degrees



actual length ratio

side / diagonal $= \phi^{-1} = \gamma$

squared ratio $= \phi^{-2} = c_0$

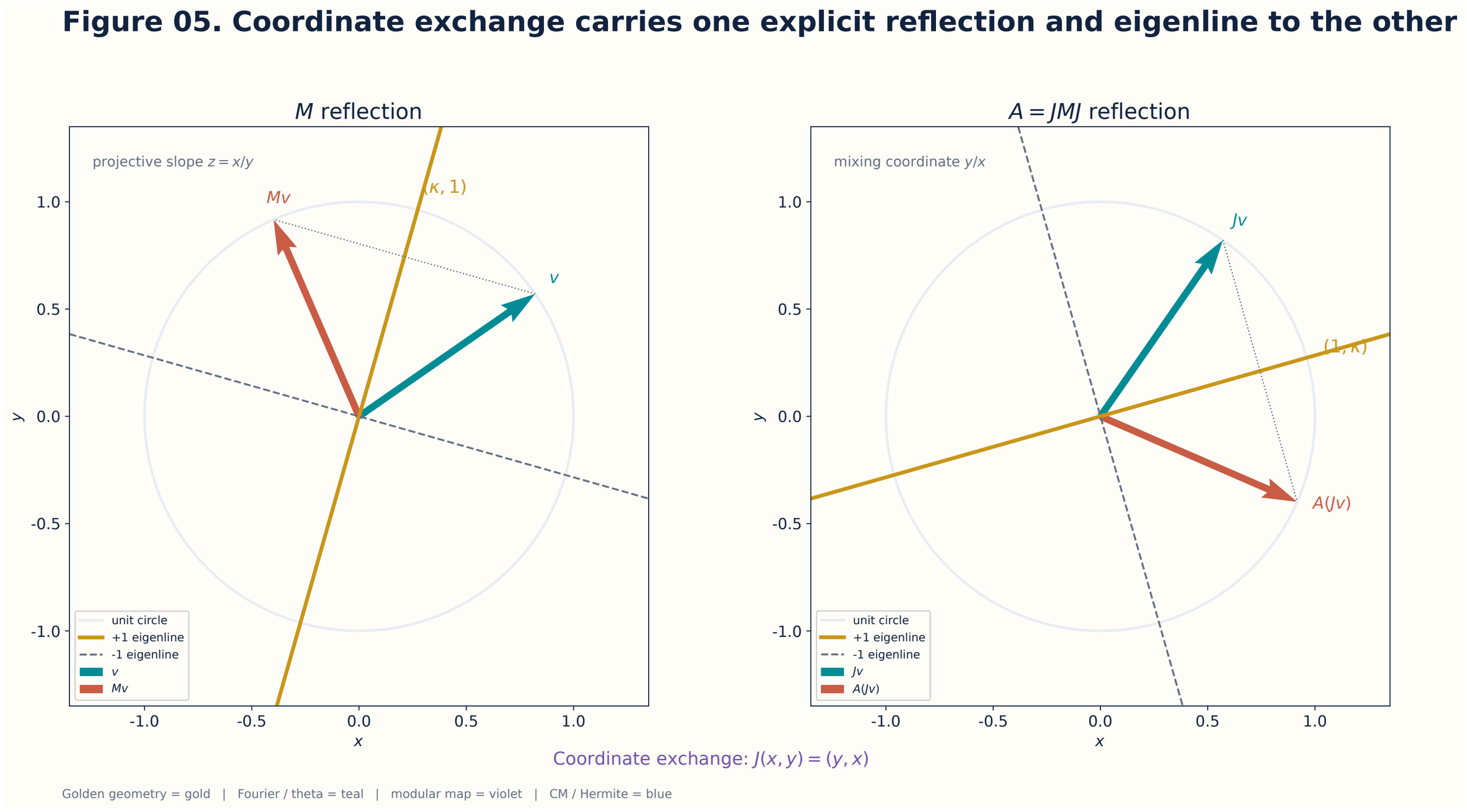
On this unit circle, the horizontal projection has value $u_0 = \cos 72^\circ$.

Golden geometry = gold | Fourier / theta = teal | modular map = violet | CM / Hermite = blue

A unit pentagon placed at angles 0, 72, 144, ... degrees. Its horizontal projection is cos(72 degrees)=u0, while the side/diagonal ratio separately gives phi^-1.

Placement cue: Section 1, after c0 and u0 are identified.

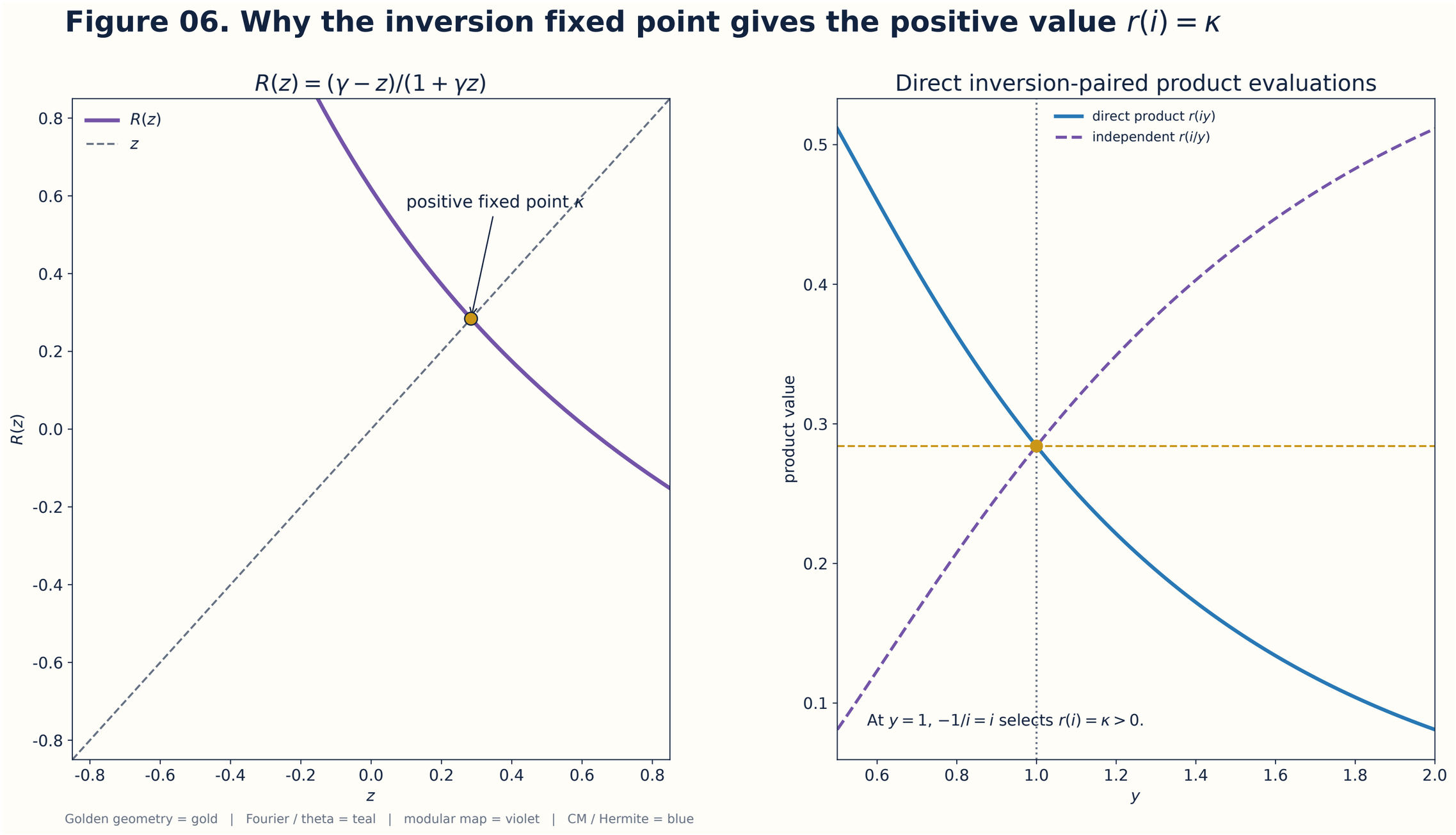
Figure 05. Coordinate exchange of reflections



M and JMJ are coordinate-exchanged plane reflections. The paired inputs v and Jv and their actual images make the exchange visible; the two slope conventions are labelled separately.

Placement cue: Section 2, alongside M, J, and A=JMJ.

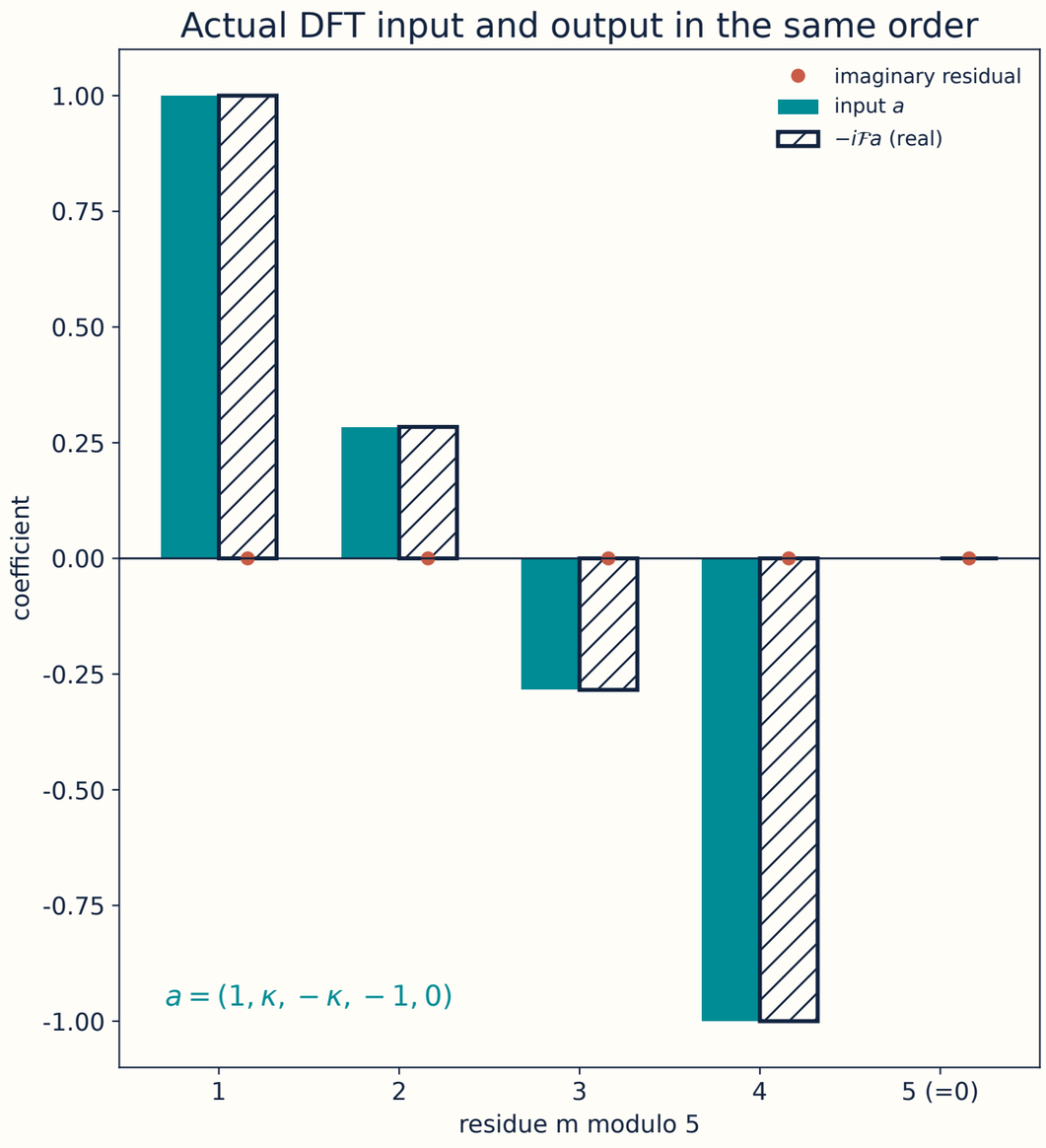
Figure 06. Why r(i)=kappa



The left panel gives the positive fixed point of the inversion map. The two curves on the right are independently evaluated product values r(iy) and r(i/y).

Placement cue: Section 2, after the inversion fixed-point argument.

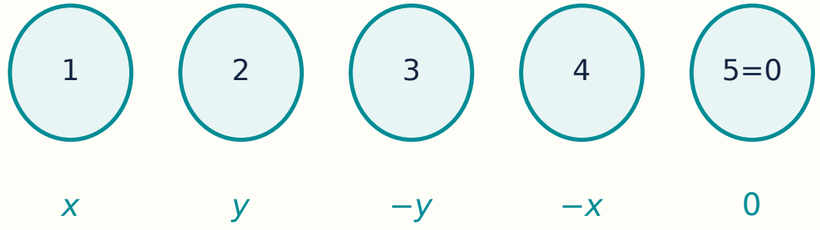
Figure 07. The odd five-periodic Fourier eigenvector, residue by residue



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Odd five-periodic coordinates

$$(a_1, a_2, a_3, a_4, a_0) = (x, y, -y, -x, 0)$$

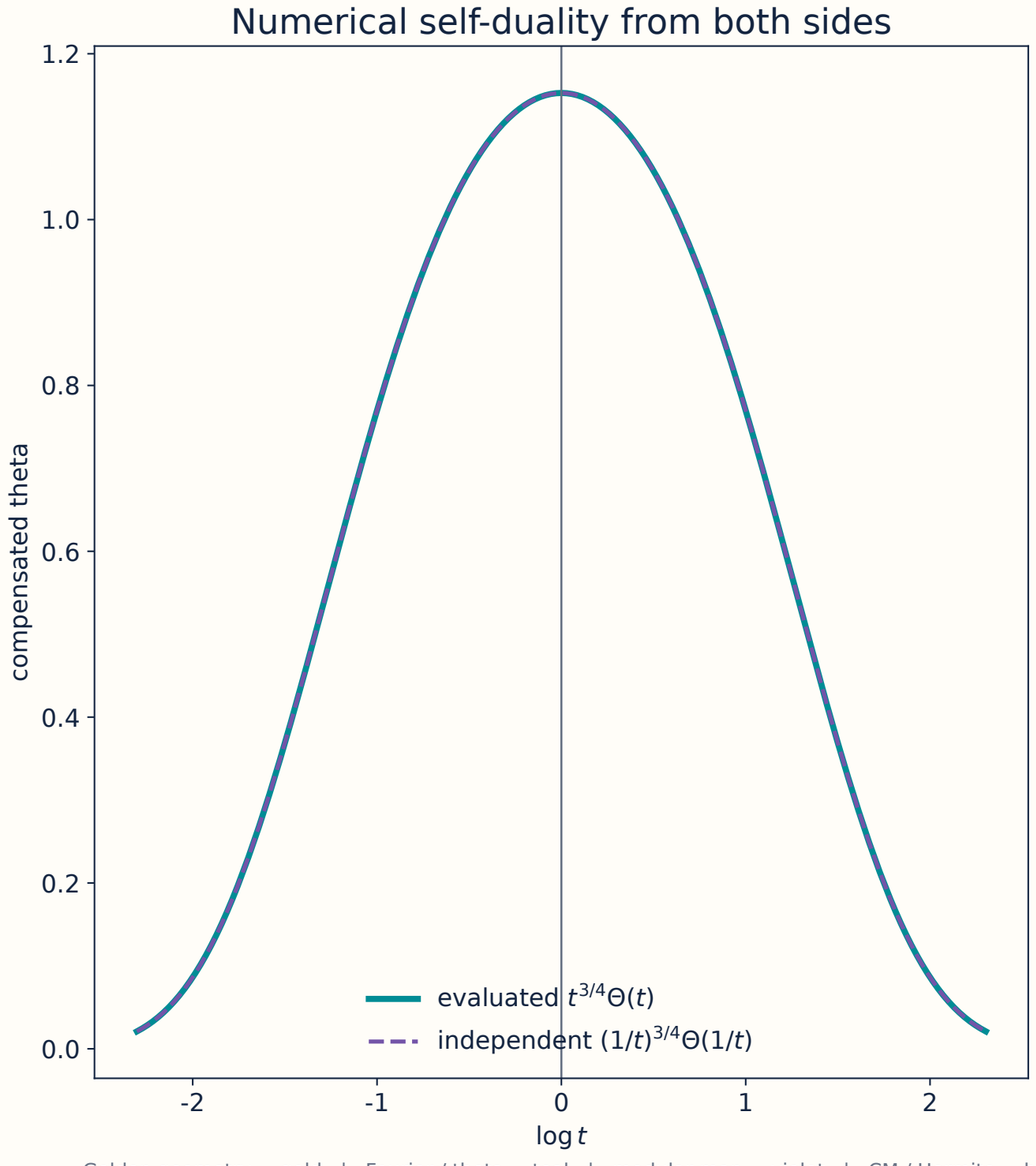


Positive-exponential normalized DFT; multiplication by $-i$ makes the odd subspace real.

Input and output are both recorded in residue order 1,2,3,4,5 (with 5 representing 0). The small imaginary residual and matching coefficients are plotted explicitly.

Placement cue: Section 2, with the finite DFT eigenvector.

Figure 08. Theta self-duality: a directly evaluated compensated symmetric profile



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Gaussian + periodic coefficients

$$\Theta_a(t) = \sum_{n \in Z} n a_n e^{-\pi n^2 t/5}$$

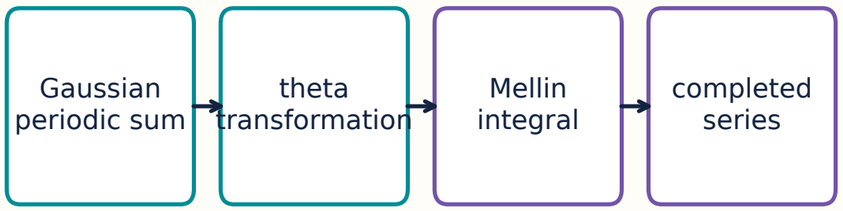
$$\Theta_a(t) = t^{-3/2}\Theta_a(1/t)$$

Both positive and negative n were summed independently for every plotted t.

Both sides of the compensated theta relation are evaluated from the Gaussian sum, including positive and negative indices, rather than constructed by reflection.

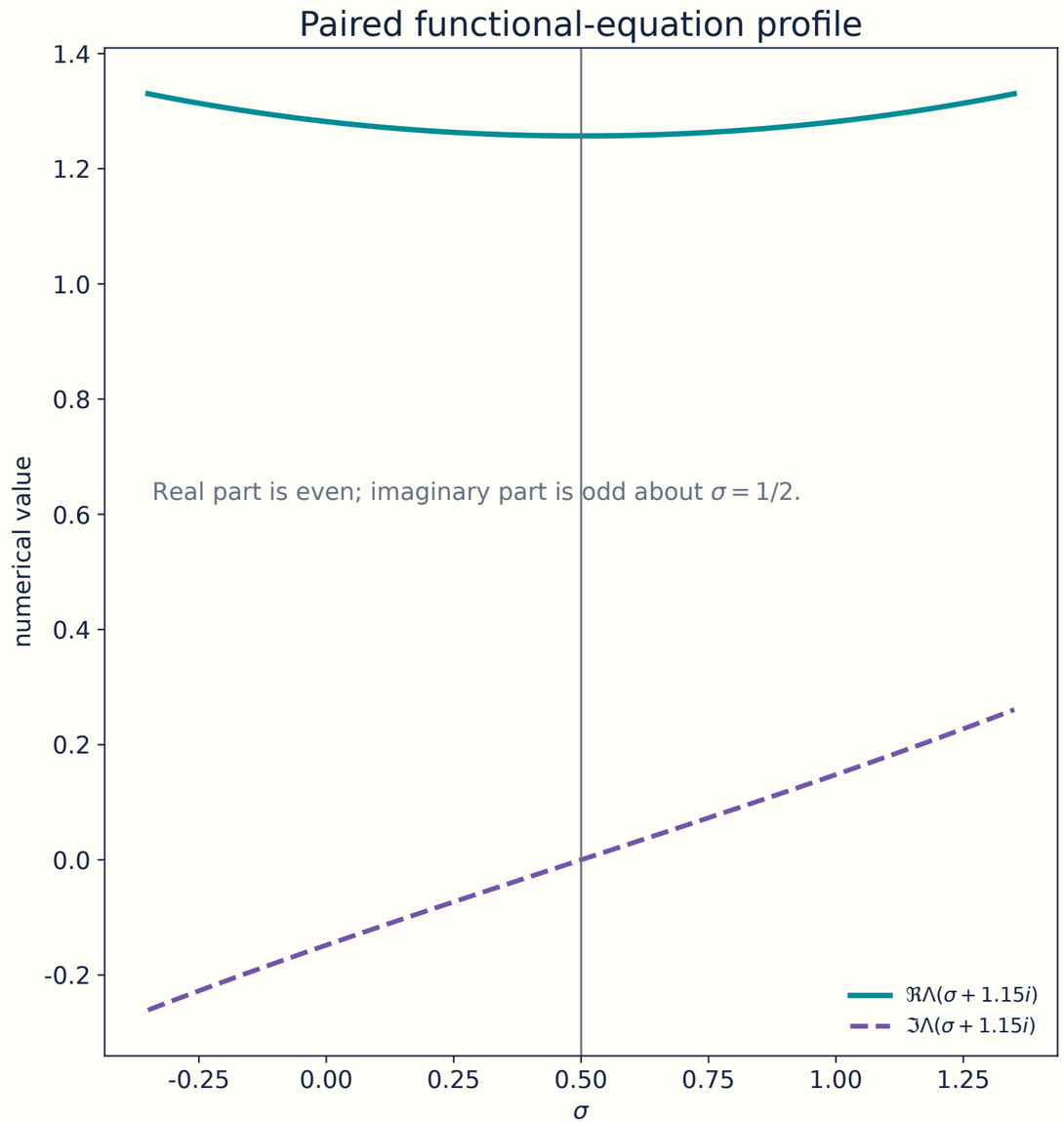
Placement cue: Section 3, after the theta construction.

Figure 09. From Gaussian sum to Mellin completion: a numerical functional-equation view



$$\Lambda(s) = (5/\pi)^{(s+1)/2}\Gamma((s+1)/2)\,5^{-s}\sum_{j=1}^{4} a_j \zeta(s, j/5)$$

For the Fourier eigenvector: $\Lambda(s) = \Lambda(1-s)$

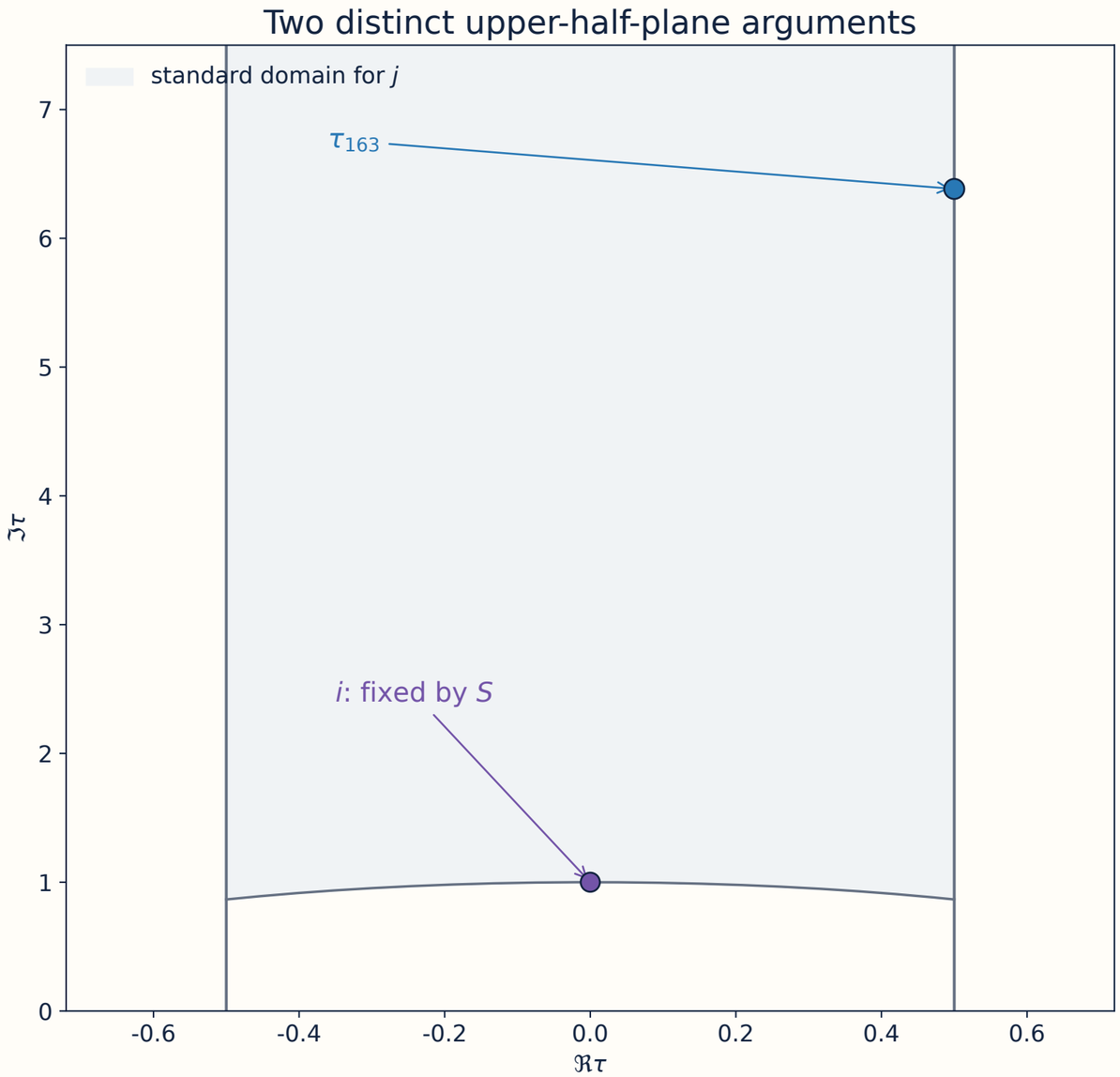


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The plate places the Gaussian-to-Mellin construction next to numerical paired completed-function values. It illustrates the stated symmetry and is not a zero search.

Placement cue: Section 3, with the Mellin completion.

Figure 10. Two CM arguments, two roles: inversion fixed point versus Hermite’s small nome



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i

$q = e^{-2\pi}$

$j(i) = 1728$

fixed point selects $r(i) = \kappa$

$\tau_{163} = (1 + i\sqrt{163})/2$

$q = -e^{-\pi\sqrt{163}},\ j = -640320^3$

integral j-value and tiny negative nome

The inversion fixed point i and tau_163 have separate jobs: r(i) selects kappa, while tau_163 supplies the integral j-value and tiny negative nome.

Placement cue: Transition to the CM comparison.

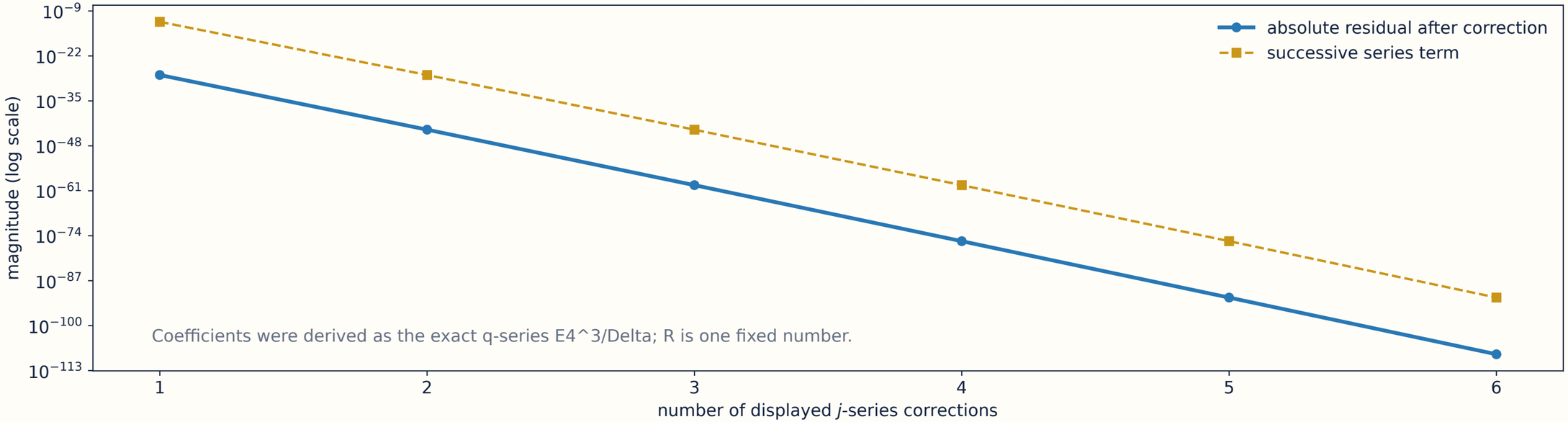
Figure 11. Hermite’s near-integer: fixed high-precision gap and the j-series correction ladder

$R = e^{\pi\sqrt{163}}$ 262537412640768743.99999999999925007

$N = 640320^3 + 744 =$ 262537412640768744

$N - R =$ 0.00000000000074992740280181431112064614366266300913729246 3

$$N - R = 196884/R - 21493760/R^2 + 864299970/R^3 + \cdots$$



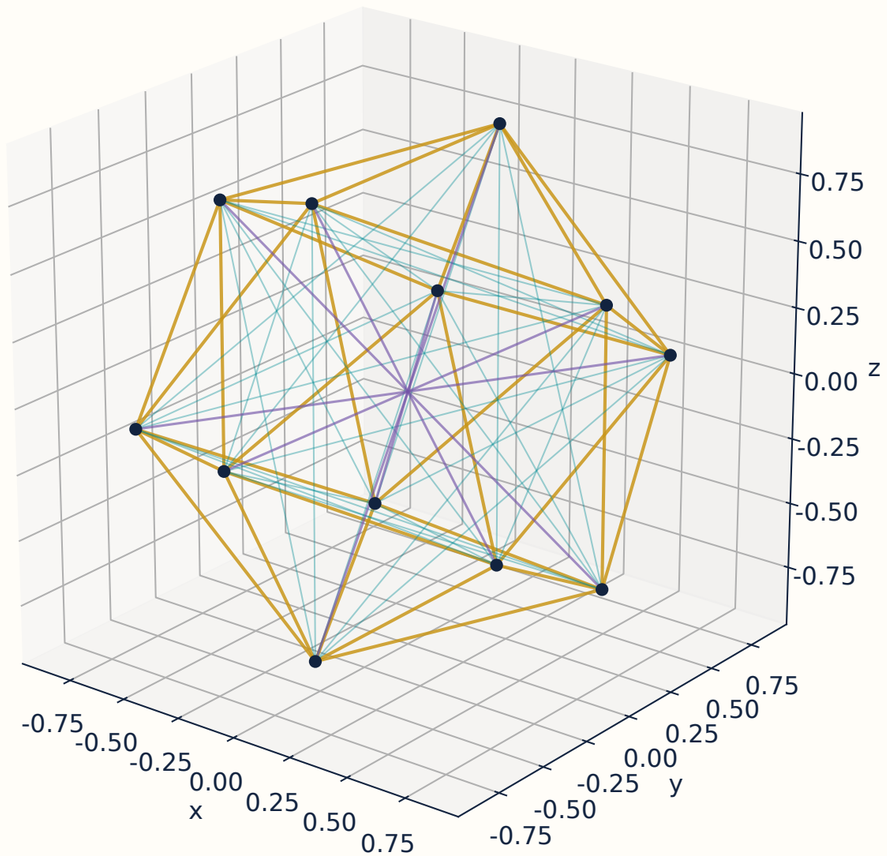
Golden geometry = gold | Fourier / theta = teal | modular map = violet | CM / Hermite = blue

At 180 decimal digits, N-R is fixed and positive. The logarithmic ladder uses the independently expanded j-series corrections 196884/R, -21493760/R^2, and onward.

Placement cue: Hermite comparison passage.

Figure 12. Context: the marked icosahedral chord ratio realizes c_0

Unit-radius marked icosahedron



All 66 unordered pairs

30 short chords

30 long chords

6 antipodal pairs

$$\frac{(\text{short})^2}{(\text{long})^2} = \frac{2 - 2/\sqrt{5}}{2 + 2/\sqrt{5}} = c_0 = \phi^{-2}$$

Golden geometry = gold | Fourier / theta = teal | modular map = violet | CM / Hermite = blue

The marked unit icosahedron supplies a Euclidean context for c0: among all 66 pairs there are 30 short, 30 long, and 6 antipodal pairs.

Placement cue: Geometry-context companion, not a CM point plot.

Sources, notation, and replay notes

Canonical source: input/MANUSCRIPT_ENDING.md sections 1-4. Context-only icosahedral source: input/GEOMETRY_CONTEXT.md section 2.

Notation: phi=(1+sqrt(5))/2; c0=phi^-2; gamma=1-c0; u0=gamma/2; kappa=r(i).

Figure 02 uses S(t,u) as written in the canonical source. Figures 06, 08, 09, and 11 are numerical renderings with saved inputs and precision settings.

The j-series coefficients used by Figure 11 are independently expanded from E4^3/Delta; the decimal gap is evaluated at 180 digits.

Replay: run build_figures.py with the arithscope Python runtime, check_figure_data.py, then build_atlas.py with the bundled ReportLab/pypdf runtime.

All figure PDFs are vector; standalone artwork is merged here rather than rasterized.

Primary references named in the canonical section remain the authoritative conventional references for the formulas shown.